

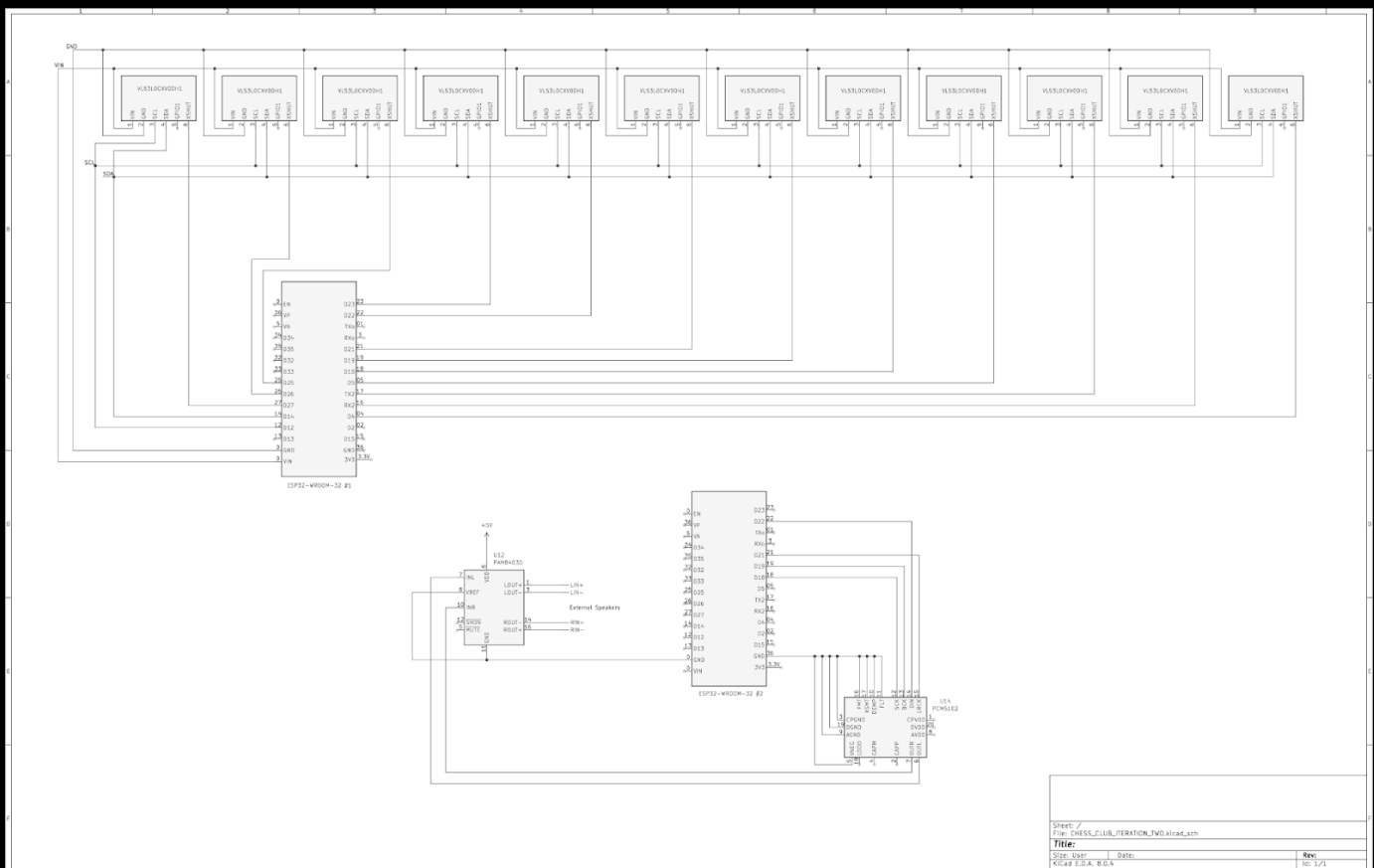
# UTOPIA ELECTRONIC DEVELOPMENT SPECIFICATIONS

## PROJECT OBJECTIVES

- Design a system of **ESP-32 microcontrollers** programmed to perform **waveform synthesis** and **audio effects** manipulated by **data** from **proximity sensors**.
- Integrate an **oscilloscope visual** for monitoring the performance of sound.
- Successfully integrate **an external sound system** for the hardware synthesizer to output sound.

## DEVELOPMENT FLOW

1. internal waveform generation to Speaker output
2. internal audio effect(s)
3. oscillator output
4. microphone input



PRELIMINARY SCHEMATICS SUBMITTED FOR PROPOSAL

## SCHEMATIC OVERVIEW

### *\* power*

- 3.3V power supply
  - VCC + GND
- 20 power (VDD) terminals to 3.3V
  - 7 ESP32s, 11 sensors, 1 microphone, 1 OLED display
- External power for subwoofer & speaker

### *\* inputs*

- **sensors**
  - VIN to 3.3V
  - GND to GND
  - SCL to ESP32 SCL
  - SDA to ESP32 SDA
  - GPIO (none)
  - XSHUT to unique GPIO on ESP32
- **microphone**

### *\* processors*

- ESP32s
- amplifiers

### *\* outputs*

- speakers
- subwoofer
- oled display

## COMPONENT SPECIFICATIONS

(x7) ESP32-ROVER-E

[documentation]([https://www.espressif.com/sites/default/files/documentation/esp32-wrover-e\\_esp32-wrover-ie\\_datasheet\\_en.pdf](https://www.espressif.com/sites/default/files/documentation/esp32-wrover-e_esp32-wrover-ie_datasheet_en.pdf))

- 2 data processors, 3 audio processors, 1 oscilloscope processor, 1 audio output/master coordinator.
- 3.3V
- 8MB PSRAM, 8MB Flash
- Wifi & Bluetooth

(x11) VL53L4CX Time-of-Flight Sensor [listing](<https://www.adafruit.com/product/5425>)

- 3.3V
- I2C communication
- SCL, SDA, GPIO, XSHUT pins

(x1) MAX9814 microphone

[documentation](<https://cdn-shop.adafruit.com/datasheets/MAX9814.pdf>)

(x2) MAX98357 Amplifier

[documentation](<https://learn.adafruit.com/adafruit-max98357-i2s-class-d-mono-amp>)

(x1) 128x64 STEMMA QT OLED display

[documentation](<https://learn.adafruit.com/monochrome-oled-breakouts>)

## REFERENCES

*A quick development platform for ESP32-A1S digital effect pedal...!*

[repository](<https://github.com/hamuro80/blackstomp>)

- Taptempo Stereo (Ping-Pong) Delay
- Stereo Chorus
- BLE Pedal
- MIDI Pedal
- Mic Mixer
- Gain Doubler

- *Distortion*

microphone and audio processing

[library](<https://github.com/sheavey/ESP32-AudioInI2S>)

## **ALLOCATION**

*ESP32 Processor #1: Data collection from ToF sensors*

*Processor #2: Data collection (...)*

*#3: Waveform synthesizer & basic manipulations*

*#4: Undecided waveform manipulation*

*#5: Undecided waveform manipulation*

*#6: Oscillator generation for OLED display*

*#7 Audio output processor & mixer for speakers*